

Pollinating *Philodendron*

The practical one-pager
Robert Wainblat

The method, with the botany left out. **Philodendron are protogynous**, and the offset is wide: on one inflorescence the female flowers are receptive **about 48 hours before** the males above them shed pollen.² No inflorescence beats its own clock. **You need two plants out of step, or pollen you saved** — which is why half this card is about pollen.

THE FOUR-DAY CYCLE

Day 1	Find and flag plump, unopened inflorescences. Gather your kit while you wait. ²
Day 2 · ~17:00	The spathe is open and the spadix is warm to the touch — reported as high as 112 °F / 44 °C . ² Receptivity and heat peak together, and the peak lasts under two hours. This is your window. Pollinate now.
Day 3	Nothing visible changes. Left alone, the spathe would close slightly. ²
Day 4	Pollen sheds from the male zone, in strings, base to apex — often in the small hours. ⁴ Collect it now. This happens once.

WHAT YOU NEED

- Sharp **craft knife**
- Soft **camel-hair brush**¹
- Rubbing alcohol
- Small airtight tube
- Silica gel / desiccant
- **Fridge** (see panel)
- Loop-lock labels
- Notebook — dates matter

READING THE RESULT

Blackened, crusty	It took. Fruit is developing. ²
Decaying	It failed. ²
A curl at the tip of a closed spathe	It opened and closed without being pollinated. It will decay. ²
Spathe dilating, then splitting off by itself	Fruit is ripe. ⁵

THE METHOD

- Work two plants, not one.** With a 48-hour offset inside every inflorescence, a cross needs a second plant flowering out of step — or pollen you stored from an earlier flowering.²
- Open the spathe. Two working methods; pick one.** Cut it away entirely with a sharp knife² — or cut away only the **top half** and leave the bottom half in place to protect the floral chamber.⁵ **The second is the more conservative**, and its reasoning is good: the spadix drops after a successful cross anyway, and the lower chamber is what carries the fruit.
- Check the stigmas before you commit.** Receptive female flowers carry a **sticky covering** — a tiny drop of fluid on each stigma tip — and that fluid is what holds the pollen.^{1,3} No fluid, no window.
- Apply to the basal third of the spadix**, where the female flowers are.² Brush or fingertips, **moistened first in the stigmatic fluid itself**.¹ Coat them generously.
- Or use the slurry method** if the chamber is hard to reach: pollen in a little sterile water, run down the sides of the spadix with an eye dropper, then tape the spathe shut.³ *The quoted “at least sixteen minutes” has no stated basis — a habit, not a rule.*
- Collect on day 4 and label it at once** with plant and date. Cut the spadix through the middle, above the flowers you pollinated, and take the pollen from the top.² **Lift it while it is still white** — it is adhesive when fresh and ambers within about **30 minutes** in air.⁴
- Then either use it now or store it — see below.**

STORING POLLEN — AN OPEN QUESTION, HONESTLY REPORTED

Three experienced sources disagree, and this card will not pretend otherwise.

McColley & Miller, 1965 — **dry** pollen kept **six weeks refrigerated at about 38 °F / 3 °C**. Their rule of thumb is worth memorising: **“if the plant will not take 38 degrees F., neither will the pollen.”**¹

LariAnn Garner — does **not** recommend freezing species whose pollen comes away **wet and stringy**; those are “likely to fail.”³

[Alicia + Nige \(@wild.about.plants\)](#) — has **frozen it repeatedly and succeeded**, including pollen chilled ten days, *then* frozen, *then* thawed, which carried an infructescence to **day 99**. They say plainly they did not expect it to work.⁵

The likely reconciler, from the same growers: pollen texture — flaky, crumbly, stringy, powdery — is **“totally dependent on species.”**⁵ So the question is not whether *Philodendron* pollen freezes, but whether **yours** does, and **nobody has tested that for any named species**. If you try it: dry, labelled, thawed still sealed — and write down what happened.

Sources. ¹ McColley & Miller 1965, *Proc. Fla. State Hort. Soc.* 78: 409–415 · ² Miller & Newton 2006, *Proc. Fla. State Hort. Soc.* 119: 425–428 · ³ Boos & Lucas, ExoticRainforest.com, relaying LariAnn Garner · ⁴ Vannini 2021, *Exotica Esoterica* · ⁵ Alicia + Nige ([@wild.about.plants](#)), Sydney, 2021–22. Photographs and figures reproduced by permission, credited to source.

What kind of knowledge this is. Almost everything here is **practice**, not measurement — no published study has tested pollen viability against storage time for any *Philodendron*. Where sources disagree, this card names them and lets them disagree.